

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
	Circles correct answer	AO1.1b	B1	$\frac{-1}{6x} + c$
	Total 1 mark			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
	Expands the bracket and obtains any correct form (PI)	AO1.1b	B1	$f'(x) = 4x^2 - 12 + \frac{9}{x^2}$ $f(x) = \int 4x^2 - 12 + \frac{9}{x^2} dx$ $= \frac{4x^3}{3} - 12x - \frac{9}{x} + c$ $f(3) = 36 - 36 - 3 + c = 2$ $c = 5$ $f(x) = \frac{4x^3}{3} - 12x - \frac{9}{x} + 5$
	Integrates at least one of 'their' terms correctly	AO1.1a	M1	
	Substitutes $x = 3$ into 'their' integrated expression and equates to 2	AO1.1a	M1	
	Obtains completely correct expression, must be explicitly stated CAO (ACF)	AO1.1b	A1	
	Total 4 marks			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
	Circles correct answer	AO1.1b	B1	$\frac{dy}{dx} = -\frac{2}{x^3}$
	Total 1 mark			

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States correct value of p	AO1.2	B1	$p = \frac{1}{2}$
(i)				
(ii)	States correct value of q	AO1.2	B1	$q = -2$
(b)	Uses valid method to find x , PI	AO1.1a	M1	$\frac{1}{2} + x = -2$
	Obtains correct x , ACF	AO1.1b	A1	$x = -2.5$
	Total 4 marks			

Q5.

	Marking Instructions	AO	Marks	Typical Solution
	States a correct integral expression (ignore limits at this stage)	AO1.1a	M1	$\text{Area} = \int_a^{2a} \left(6x^2 + \frac{8}{x^2} \right) dx$
	Integrates at least one term correctly	AO1.1b	A1	$= \left[2x^3 - \frac{8}{x} \right]_a^{2a}$
	Substitutes $2a$ and a into 'their' integrated expression	AO1.1a	M1	$= \left(16a^3 - \frac{4}{a} \right) - \left(2a^3 - \frac{8}{a} \right)$
	States correct final answer with terms collected FT correct substitution into incorrect integral provided both M1 marks awarded	AO1.1b	A1F	$= 14a^3 + \frac{4}{a}$
	Total 4 marks			

Q6.

	Marking Instructions	AO	Marks	Typical Solution
	Circles correct answer	AO1.1b	B1	$-\frac{1}{x\sqrt{x}}$
	Total 1 mark			

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Separates variables, at least one side correct.	AO3.1a	M1	$3\sqrt{x} \frac{dx}{dt} = 8 \sin 2t$

				$\int 3\sqrt{x} \, dx = \int 8 \sin 2t \, dt$ $\int 3x^{\frac{1}{2}} dx = \int 8 \sin 2t \, dt$ $2x^{\frac{3}{2}} = -4 \cos 2t (+c)$
	Obtains correct separation PI	AO1.1b	A1	
	integrates 'their' expressions, at least one of 'their' sides correct	AO1.1a	M1	
	Obtains correct integral (condone missing + c) CAO	AO1.1b	A1	
	Substitutes initial conditions, to find + c.	AO3.1b	M1	$2 \times (0)^{\frac{3}{2}} = -4 \cos(2 \times 0) + c$ $c = 4$
	Obtains a correct solution ACF	AO1.1b	A1	$x^{\frac{3}{2}} = 2 - 2 \cos 2t$
	Obtains correct solution of the form $x = f(t)$	AO2.5	A1	$x = (2 - 2 \cos 2t)^{\frac{2}{3}}$
(b)	Obtains correct max height, in cm Award FT from correct substitution into incorrect equation $x = f(t)$ but only if all three M1 marks have been awarded, must have correct units.	AO3.4	A1F	$\text{max height} = 4^{\frac{2}{3}} = 252 \text{ cm}$
	Total 8 marks			

Q8.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving  and a constant of proportionality.	AO3.3	M1	$\frac{dA}{dt} \propto A$ $\Rightarrow \frac{dA}{dt} = kA$ $\Rightarrow \int \frac{1}{A} dA = \int k \, dt$ $\Rightarrow \ln A = kt + c$ $\Rightarrow A = e^{kt+c}$ $\Rightarrow A = Be^{kt} \quad \mathbf{AG}$
	Separates variables	AO1.1a	M1	
	Integrates both of 'their'	AO1.1b	A1F	

	sides correctly			
	Constructs a rigorous mathematical argument that supports use of the given model. AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
(b)	States correct value of B	AO1.1b	B1	$B = 0.25$ or $B = \frac{1}{4}$
(i)				
(ii)	Uses $t = 20$ and $A = 0.5$ to find k	AO3.1b	M1	When $t = 20$, $A = 0.5$ $\Rightarrow 0.5 = 0.25e^{20k}$ $\Rightarrow 20k = \ln 2$ $\Rightarrow k = \frac{1}{20} \ln 2$ $\Rightarrow A = \frac{1}{4} (e^{\ln 2})^{\frac{t}{20}}$ $\Rightarrow A = 2^{-2} \times 2^{\frac{t}{20}}$ $A = 2^{\frac{t}{20}-2}$ AG
	Finds correct value of k	AO1.1b	A1	
	Substitutes 'their' k to get A in terms of t	AO1.1a	M1	
	Constructs rigorous and convincing argument to show $A = 2^{\frac{t}{20}-2}$ Using correct notation throughout. AG	AO2.1	R1	
(iii)	Uses the model to set up correct equation and attempt to find t	AO3.4	M1	$2\pi = 2^{\frac{t}{20}-2}$ $t = 93.03$ days
	Finds correct value of t	AO1.1b	A1	

(c)	States any sensible and relevant limitation of the model that is specified in terms of the pond, area, weed, rate of change or time.	AO3.5b	E1	Model predicts that the area of weed will increase without limit and this is not possible since the area of the pond is 4π
(d)	Any sensible and relevant refinement to the model that is specified in terms of	AO3.5c	E1	Introduce a limiting factor such as fish eating weed or rate of growth

	the pond, area, weed, rate of change or time			decreases as surface area covered
	Total 13 marks			

Q9.

(a) $\sqrt{x} = x^{0.5}$

$\sqrt{x} = x^{0.5}$ or $\sqrt{x} = x^{\frac{1}{2}}$ seen or used

B1

$$\frac{12 + x^2 \sqrt{x}}{x} = \frac{12 + x^{2.5}}{x}$$

$$= 12x^{-1} + x^{1.5}$$

$12x^{-1}$ or $p = -1$

B1

$x^{1.5}$ or $q = \frac{3}{2}$ ($=1.5$)

B1

3

(b) (i) $\frac{dy}{dx} = -12x^{-2}$

Ft on c's p only if c's p is a negative integer

B1F

$+ 1.5x^{0.5}$

Ft on c's q only if c's q is a pos non-integer

B1F

2

(ii) When $x = 4$, $y = 11$

B1

When $x = 4$, $\frac{dy}{dx} = \frac{-12}{16} + 3 = \frac{9}{4}$

Attempt to find $\frac{dy}{dx}$ when $x = 4$ PI

M1

Gradient of normal = $-\frac{4}{9}$

$m \times m' = -1$ used

m1

Eqn of normal: $y - 11 = -\frac{4}{9}(x - 4)$

ACF eg $4x + 9y = 115$

A1

4

(iii) At St Pt $\frac{dy}{dx} = -12x^{-2} + 1.5x^{0.5} = 0$

Equating c's $\frac{dy}{dx}$ to zero

M1

$\Rightarrow x^2 x^{0.5} = 8, \Rightarrow x^{\frac{5}{2}} = 8 \Rightarrow x = 8^{\frac{2}{5}}$

A correct eqn in the form $x^n = c$ or $x = c^{\frac{1}{n}}$ correctly obtained.

A1

$\Rightarrow x = (2^3)^{\frac{2}{5}} \Rightarrow x = 2^{\frac{6}{5}}$

CSO $x = 2^{\frac{6}{5}}$. All working must be correct and in an exact form.

If 'x = 0' also appears then A0 CSO

A1

3

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[12]

Q10.

(a) $\sqrt[4]{x^3} = x^{\frac{3}{4}}$

Accept $k = \frac{3}{4}$ OE

B1

1

(b) $\frac{1-x^2}{\sqrt[4]{x^3}} = \frac{1}{\sqrt[4]{x^3}} - \frac{x^2}{\sqrt[4]{x^3}} = x^{-k} - \frac{x^2}{\sqrt[4]{x^3}} \quad \left[\text{or } \frac{1}{\sqrt[4]{x^3}} - x^{2-k} \right]$

Split followed by at least one correct index law used to remove denominator.

M1

$= x^{-\frac{3}{4}} - x^{\frac{5}{4}}$

If incorrect, ft on c's non-integer k value answer to part (a), provided M1 has been awarded. Accept answer given in

form of values for p and q.

A1F

2

[3]

Q11.

(a) (i) $(p =) 3$

B1

1

(ii) $(q =) -3$

If not correct, ft on $-p$

B1F

1

(iii) $(r =) \frac{1}{2}$
OE

B1

1

(b) $2^{\frac{1}{2}} \times 2^x = 2^{-3} \Rightarrow 2^{\frac{1}{2}+x} = 2^{-3}$

Using a law of indices or logs correctly to combine at least two of the powers of 2 PI

M1

$$\Rightarrow x = -3\frac{1}{2}$$

If not correct, ft on $x = q - r$ provided method Shown A1F

2

[5]

Q12.

(a) $(2 + x^2)^3$

$$= [(2)^3] + 3(2)^2(x^2) + 3(2)(x^2)^2 [+ (x^2)^3]$$

For either (1), 3, 3, (1) OE unsimplified

or $\binom{3}{1} 2^2 x^2 + \binom{3}{2} 2(x^2)^2$ OE. PI

M1

$$p = 3(2)^2 = 12$$

AG Be convinced. Condone left as $12x^2$

A1

$$q = 6$$

Accept left as $6x^4$

B1

3

(b) (i)
$$\int \frac{(2+x^2)^3}{x^4} dx = \int x^{-4} (8 + 12x^2 + qx^4 + x^6) dx$$

or
$$\int \left(\frac{8}{x^4} + \frac{12}{x^2} + q + x^2 \right) dx$$

Uses (a) and either an indication that $\frac{1}{x^n} = x^{-n}$
in a product PI or cancelling to get at
least 3 correct ft terms

M1

$$\int (8x^{-4} + 12x^{-2} + q + x^2) dx$$

Ft on c's non-zero q. PI by next line in solution

A1F

$$= \frac{8x^{-3}}{-3} + \frac{12x^{-1}}{-1} + qx + \frac{x^3}{3} + c$$

Correct integration of either $8x^{-4}$ or $12x^{-2}$;
accept unsimplified

M1

Correct integration of both $8x^{-4}$ and $12x^{-2}$;
accept unsimplified coefficients

A1

$$= \dots\dots\dots + 6x + \frac{x^3}{3} + c$$

$$\left(= -\frac{8}{3}x^{-3} - 12x^{-1} + 6x + \frac{x^3}{3} + c \right)$$

For $6x + \frac{x^3}{3} + c$ simplified. The only ft is "6"
replaced by c's value for q where q is a
non-zero integer.

B1F

5

(ii)
$$\int_1^2 \frac{(2+x^2)^3}{x^4} dx = \left\{ -\frac{8}{3}(2)^{-3} - 12(2)^{-1} + 6(2) + \frac{2^3}{3} \right\} -$$

$$\left\{ -\frac{8}{3}(1)^{-3} - 12(1)^{-1} + 6(1) + \frac{1^3}{3} \right\}$$

Dealing correctly with limits; $F(2)-F(1)$ (must
have attempted integration to get F ie c's F is
not just the integrand)

M1

$$= \left(-\frac{1}{3} - 6 + 12 + \frac{8}{3} \right) - \left(-\frac{8}{3} - 12 + 6 + \frac{1}{3} \right)$$

$$= 16\frac{2}{3}$$

OE **exact** answer eg $50/3$.
NMS scores 0

A1

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[10]

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Q13.

(a) $\sqrt{x^5} = x^{\frac{5}{2}}$

Accept $k = 2.5$

B1

1

(b) $\int (7\sqrt{x^3} - 4) dx = \frac{7}{3.5} x^{3.5} - 4x(+c)$

Index ' k ' raised by 1 in integrating x^k

M1

1st term correct follow through on non- integer k

A1F

For $-4x$ as integral of -4

B1

3

- (c) $y = 2x^{3.5} - 4x + c$ (*)
 $y = c$'s answer to (b) with '+ c'
 ('y =' PI by next line)

B1F

When $x = 1, y = 3$ $3 = 2 - 4 + c$
Subst. (1, 3) in attempt to find constant of integration

M1

$y = 2x^{3.5} - 4x + 5$
*Accept $c = 5$ after correct eqn * which must include 'y ='*
Coefficients must be tidied

A1

3

[7]

Q14.

(a) $\sqrt{x} = x^{\frac{1}{2}}$

PI

$$\frac{x^3 + \sqrt{x}}{x} = \frac{x^3}{x} + \frac{\sqrt{x}}{x} = x^2 + x^{-\frac{1}{2}}$$

Accept $p = 2; q = -\frac{1}{2}$

B1

B1;B1

3

(b) (i) $\frac{dy}{dx} = 2x - \frac{1}{2}x^{-\frac{3}{2}}$

Reduces both powers by 1

M1

ACF

A1

2

(ii) When $x = 1, y = 2$

PI if not stated explicitly eg the '2' may appear in the correct posn. in later eqn.

B1

When $x = 1, \frac{dy}{dx} = 2 - \frac{1}{2} = \frac{3}{2}$

Attempt to find $\frac{dy}{dx}$ When $x = 1$ PI

M1

Gradient of normal = $-\frac{2}{3}$

*-1/ (c's value of dy/dx when $x = 1$)
either stated as the gradient of the normal or used as the gradient in the equation of the normal*

m1

Equation of normal: $y - 2 = -\frac{2}{3}(x - 1)$

*Only ft on c's $\frac{dy}{dx}$ in part (b)(i)
ACftF*

A1F

4

(c) (i) $\frac{d^2y}{dx^2} = 2 + \frac{3}{4}x^{-\frac{5}{2}}$

Reduces both powers by 1.

M1

Ft on (b)(i) provided at least one power to be differentiated is both negative and fractional

A1F

2

(ii) (Since $x > 0,$) $\frac{d^2y}{dx^2} > 0$

For a maximum point $\frac{d^2y}{dx^2}$ is **not** positive so C has no maximum points

*E1 for attempt to find the sign of $\frac{d^2y}{dx^2}$;
either in general terms or at the pt(s)
where c 's $dy/dx = 0$ or the remaining E mark*

*a correct justification for why $\frac{d^2y}{dx^2} > 0$ and also
a full correct concluding statement must be made.*

E2,1,0

2

[13]

Q15.

(a) (i) $\log_a 40$

Accept ' $k = 40$ '

B1

1

(ii) $\log_a 8$

Accept ' $k = 8$ '

B1

1

(iii) $\log_a 125$

Accept ' $k = 125$ ' but not ' $k = 5^3$ '

B1

1

(b) $\log_{10} \{(1.5)^{3x}\} = \log_{10} 7.5$

*Correct statement having taken logs of
both sides of $(1.5)^{3x} = 7.5$ OE PI
or $3x = \log_{1.5} 7.5$ seen*

M1

$3x \log_{10} 1.5 = \log_{10} 7.5$

$\log 1.5^{3x} = 3x \log 1.5$ OE

m1

$x = \frac{\lg 7.5}{3 \lg 1.5} = 1.65645 \dots = 1.656$ to 3 dp

Both method marks must have been
awarded with clear use of logarithms seen

A1

3

(c) $\log_2 p = m \Rightarrow p = 2^m; \log_8 q = n \Rightarrow q = 8^n$

Either $p = 2^m$ or $q = 8^n$ seen or used

M1

$p = 2^m$ and $q = 2^{3n}$

Writing $8^n = 2^{3n}$ and having $p = 2^m$

m1

$pq = 2^m \times (2^3)^n = 2^m \times 2^{3n}$ so $pq = 2^{m+3n}$

Accept $y = m + 3n$

A1

3

[9]

Q16.

(a) $(n =) -4$

Accept x^{-4}

B1

1

(b) $\left(1 + \frac{3}{x^2}\right)^2 = 1 + \frac{6}{x^2} + \frac{9}{x^4}$

Apply ISW after B2 stage

(B1 if correct but unsimplified seen)

B2,1,0

2

(c) $\int \left(1 + \frac{3}{x^2}\right)^2 dx = x - 6x^{-1} - 3x^{-3} + c$

*At least one power of x correctly obtained
in the integration of an expansion*

M1

A2 terms correct and "+ c"

(A1F two terms in x correct ft on

*expansion provided integrating x to a
negative power)*

A2, 1, 0

3

(d)
$$\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 dx = \left[x - \frac{6}{x} - \frac{3}{x^3}\right]_1^3$$

$$= \left(3 - \frac{6}{3} - \frac{3}{27}\right) - \left(1 - 6 - 3\right)$$

*Dealing correctly with limits; $F(3) - F(1)$
 (must have attempted integration to get F)*

M1

$$= 8\frac{8}{9}$$

CSO;

OE provided value is **exact**, eg $\frac{80}{9}, \frac{240}{27}$;
 ISW dec value after exact value seen
 NMS scores 0/2

A1

2

[8]

Q17.

(a) (i) $\sqrt{125} = \sqrt{25 \times 5} = 5\sqrt{5}$

OE eg $\sqrt{125} = \sqrt{5^3}$ or $5^{1.5}$ seen

M1

$$5^p = \sqrt{125} \Rightarrow p = 1.5$$

Correct value of p must be explicitly stated

A1

Alternative:

$$p \log 5 = \frac{1}{2} \log 125$$

2

OE eg $p \log 5 = \log 11.18$

or eg $p = \log_5 \sqrt{125}$

(M1)

$$p \log 5 = \frac{3}{2} \log 5 \Rightarrow p = \frac{3}{2}$$

Correct
explicitly stated

value of p must be

(A1)

(ii) $5^{2x} = \sqrt{125} = 5^{\frac{3}{2}} \Rightarrow x = 0.5p = 0.75$

Must be $0.5 \times c$'s value of p

SC: $x = 0.75$ with c 's ans (a)(i) $5^{1.5}$ scores B1F

B1F

1

(b) $3^{2x-1} = 0.05$

$(2x - 1)\log 3 = \log 0.05$

Take logs of both sides and use 3rd law of
logs. PI eg by $2x - 1 = \log_3 0.05$ seen

M1

$$x = \frac{\log_{10} 0.05}{2\log_{10} 3} + \frac{1}{2}$$

Correct rearrangement to $x = \dots$ PI

m1

$= -0.8634(165\dots) = -0.8634$ to 4dp

Condone > 4 dp. Must see logs clearly
used in solution, so NMS scores 0/3

A1

3

(c) $\log_a x = 2(\log_a 3 + \log_a 2) - 1$

$= 2 \log_a (3 \times 2) - 1$

A valid law of logs used

M1

$= \log_a (6^2) - 1$

Another valid law of logs used

M1

$= \log_a 36 - \log_a a$

$\log_a a = 1$ quoted or used

or $\log_a \frac{x}{k} = -1 \Rightarrow \frac{x}{k} = a^{-1}$ OE

B1

$$\log_a x = \log_a \left(\frac{36}{a} \right) \Rightarrow x = \frac{36}{a}$$

CSO Must be $x = \frac{36}{a}$ or $x = 36a^{-1}$

A1

4

[10]

Q18.

(a) $\sqrt{x^3} = x^{\frac{3}{2}}$

OE; accept 'k = 1.5'

B1

1

(b) (i) $\frac{dy}{dx} = 2x - \frac{3}{2}x^{\frac{1}{2}}$

At least one index reduced by 1 and no term of the form $\sqrt{ax^2}$.

M1

For $2x$

B1

For $-1.5x^{0.5}$. Ft on ans (a) non-integer k

A1F

3

(ii) When $x = 4$, $y = 8$

B1

$y'(4) = ;$

Attempt to find $\frac{dy}{dx}$ when $x = 4$

M1

$= 2(4) - 1.5(\sqrt{4}) = 5$

Ft on one earlier error provided noninteger powers in (a) and (b) (i)

A1F

Tangent: $y - 8 = 5(x - 4)$

$y - y(4) = y'(4)[x - 4]$ OE

m1

$$y = 5x - 12$$

CSO; must be $y = 5x - 12$

A1

5

[9]

Q19.

(a) $r = 16 \div 20 = 0.8$

OE

B1

1

(b) $\frac{a}{1-r} = \frac{20}{1-0.8}$

OE Using a correct formula with $a = 20$ or
 $r = c$'s 0.8

M1

$= 100$

ft on c's value of r provided $|r| < 1$

A1F

2

(c) $\{S_{20}\} = \frac{a(1-r^{20})}{1-r}$

OE Using a correct formula with $n = 20$

M1

$= 100(1 - 0.8^{20}) = 98.847\{07..\}$

Condone > 3dp

A1

2

(d) n th term $= 20 r^{n-1} = 20(0.8)^{n-1}$

ft on c's r . Award even if 16^{n-1} seen

M1

$= 20 \times 0.8^{-1} \times 0.8^n$

$= 25 \times 0.8^n$

CSO; AG

A1

2

[7]

Q20.

(a) (i) $\{p \Rightarrow\} 2$

Condone '64 = 8²'

B1

(ii) $\{q \Rightarrow\} -2$

Ft on '− p' if q not correct

B1ft

(iii) $\{r \Rightarrow\} 0.5$

Condone ' $\sqrt{8} = 8^{0.5}$ '

B1

3

(b) $\frac{8^x}{8^{0.5}} = 8^{-2} \Rightarrow 8^{x-0.5} = 8^{-2}$

OE

Using parts (a) & valid index law to stage

$8^c = 8^d$ (PI)

M1

$\Rightarrow x - 0.5 = -2$ $x = -1.5$

Ft on c's (q + r) if not correct

(Accept correct answer without working)

A1ft

2

ALT: $\log 8^x = \log k$, $x \log 8 = \log k$; $x = -1.5$

(M1 A1)

[5]

Q21.

(a) (i) x^2

B1

1

(ii) $x^{\frac{1}{2}} = \sqrt{x}$

Accept either form

B1

1

(iii) x^3

B1

1

(b) (i) $\int 3x^{\frac{1}{2}} dx = \frac{3}{\frac{3}{2}} x^{\frac{3}{2}} \{+c\}$

Index raised by 1

M1

Simplification not yet required

A1

$$= 2x^{\frac{3}{2}} + c$$

Need simplification and the + c OE

A1

3

(ii) $\int_1^9 3x^{\frac{1}{2}} dx = (2 \times 9^{\frac{3}{2}}) - (2 \times 1^{\frac{3}{2}})$

F(9) – F(1), where F(x) is candidate's answer to (b)(i) [or clear recovery]

M1

$$= 52$$

Ft on (b)(i) answer of form $kx^{1.5}$ i.e. $26k$

A1ft

2

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[8]

Q22.

(a) $\sqrt{x} = x^{\frac{1}{2}}$

Accept $k = 0.5$

B1

1

(b) $\sqrt{x}(x-1) = x^{\frac{1}{2}}x - x^{\frac{1}{2}} = x^{\frac{3}{2}} - x^{\frac{1}{2}}$

M1

Accept $p = 1.5, q = 0.5$

A1

2

(c) $\int \sqrt{x}(x-1) dx = \frac{x^{2.5}}{2.5} - \frac{x^{1.5}}{1.5} (+c)$
Increases a power of x by 1

M1

ft non-integer p

A1ft

ft non-integer q

A1ft

3

(d) $\int_1^2 dx = \left(\frac{2^{2.5}}{2.5} - \frac{2^{1.5}}{1.5} \right) - \left(\frac{1}{2.5} - \frac{1}{1.5} \right)$
Limits; $F(2) - F(1)$

M1

$\dots = \left(\frac{4\sqrt{2}}{2.5} - \frac{2\sqrt{2}}{1.5} \right) - \left(\frac{1}{2.5} - \frac{1}{1.5} \right)$
Fractional powers to surds

m1

$\left(\frac{24\sqrt{2}}{15} - \frac{20\sqrt{2}}{15} \right) - \left(-\frac{4}{15} \right) = \text{pr. ans}$
CSO AG (be convinced)

A1

3

[9]

Q23.

(a) (i) $\sqrt{x} = x^{\frac{1}{2}}$
Accept $p = 0.5$

B1

1

(ii) $\int \sqrt{x} dx = \frac{x^{1.5}}{1.5} \{+c\}$
Index raised by 1

M1

Correct ft on

p. Condone missing '+c'

A1ft

2

(iii) $\text{Area} = \int_1^4 \sqrt{x} \, dx$
Limits 1 and 4 PI

B1

$$\dots\dots = \frac{4^{1.5}}{1.5} - \frac{1}{1.5}$$

$$F(4) - F(1)$$

M1

$$= \frac{14}{3}$$

Accept 4.66 or better

A1

3

(b) (i) $y = x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$
Index (p - 1) ft

M1

When $x = 4$, $y'(4) = 0.25$

Attempt to find $y'(4)$

M1

When $x = 4$, $y = 2$

B1

Equation of tangent: $y - 2 = \frac{1}{4} (x - 4)$
accept other forms

A1

4

(ii) When $x = 0$, $y = 1$ $B(0, 1)$
*Subs $x = 0$ **and** then $y = 0$ into equation of tangent. PI*

M1

When $y = 0, x = -4 \quad A(-4, 0)$

*Correct ft y_B and x_A
(may be awarded as part of area calculation)*

A1ft

$$\text{Area} = 0.5(1)(4) = 2$$

ft wrong sloping tangent and max of 1 further slip. Final answer must be +ve

A1ft

3

(c) Translation

'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

*Accept equivalent in words provided linked to 'translation/move/shift'
(Note: B0B1 is possible)*

B1

2

(d) $h = 1$

PI

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B1

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$$\text{Integral} = h/2 \{ \dots \}$$

$$\{ \dots \} = f(1) + 2[f(2) + f(3)] + f(4)$$

OE summing of areas of the three traps

M1

$$\{ \dots \} = 0 + 2(1 + \sqrt{2}) + \sqrt{3}$$

Condone 1 numerical slip

A1

$$\text{Integral} = \frac{1}{2} \{ 2(1 + 1.414\dots) + 1.732 \}$$

$$\text{Integral} = 0.5 \times 6.560\dots = 3.28 \text{ to 3sf}$$

Must be 3.28

A1CAO

4

[19]

Q24.

(a) = $x^5 - x^{-3}$

One power correct

M1

Accept $p = 5, q = -3$

A1

2

(b) (i) $f'(x) = 5x^4$

ft on $px^p - 1$

B1ft

$+3x^{-4}$
ft on $-qx^q - 1$ provided $q < 0$

B1ft

2

(ii) $f'(x) \left\{ = 5x^4 + \frac{3}{x^4} \right\} > 0$

M1 Considers sign of $f'(x)$; a statement " $f'(x) > 0$ OE" with 'f increasing'.

M1

$\Rightarrow f$ is increasing {function}

*A1 needs $f(x)$ of the form $ax^4 + \frac{b}{x^4}$,
 where a and b both > 0 and no incorrect
 statements based on $f'(x)$ at different values of x*

A1

2

(c) At (1, 0), $f'(1) = 5 + 3 = 8$

Attempts to find $f'(1)$

M1

$\Rightarrow$ grad. of normal = $-\frac{1}{8}$

Use of $m \times m' = -1$ PI

m1

ft on wrong $f'(x)$

A1ft

3

[9]



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